AlumniConnect – Student & Alumni Mentoring Portal

Test Case Specification and Results Report

Item	Detail
Project	Alumni-Mentoring-Portal (EDI Lab B batch)
Repository	https://github.com/anuragpatil1729/Alumni-Mentoring-Portal
Technology	Java 17+, Maven, JDBC, MySQL / MariaDB, Vanilla HTML/CSS/JS frontend
Test framework	JUnit 5 (org.junit.jupiter); run with: cd backend && mvn clean test
Source of test cases	11 JUnit test files supplied (backend/src/test/java/com/alumni)
Test cases in main set	145 (TC-001 to TC-145) across 9 test classes
Additional cases (Appendix A)	24 (TC-146 to TC-169) across 2 further test classes
Report date	22 October 2026

1. Execution Status Notes

Status column: “Pass*” marks the 87 tests the repository README reports as passing (Tests run: 87, Failures: 0, Errors: 0, Skipped: 0, BUILD SUCCESS). “Not run” marks tests that were not part of that documented run; their results are not recorded in the repository.

These tests need a live MySQL database and the Java backend, which were not available while this report was prepared, so no test was re-executed here. After running mvn clean test, update the Status column from the Surefire report (backend/target/surefire-reports).

2. Summary by Test Class

No.	Test Class	Module Covered	Tests	Result
1	MentorshipRequestTest	Mentorship Request Lifecycle	23	23 Pass*
2	ValidationTest	Input Validation, Registration & Authentication	28	28 Pass*
3	SearchAlgorithmsTest	Mentor Search Algorithms (Linear, Quicksort, Binary)	19	19 Pass*
4	ProfileTest	Profile Management & Avatar	13	13 Pass*
5	JDBCTest	Database Connectivity & Transactions	4	4 Pass*

No.	Test Class	Module Covered	Tests	Result
6	ConflictDetectionTest	Session Schedule Conflict Detection	9	Not run
7	SessionHistorySortTest	Session History Sorting Algorithms	7	Not run
8	LoginBoundaryTest	Login Boundary & Security Testing	26	Not run
9	AuthProfileIntegrationTest	Auth & Profile End-to-End Integration	16	Not run
	TOTAL (main set)		145	87 Pass*, 58 Not run

Breakdown by Test Type (main set)

Type	Count
Unit	41
API	28
Boundary	19
Service	17
Integration	17
Security	9
Validation	7
DB	4
DAO	3

3. Detailed Test Cases (TC-001 to TC-145)

3.1 MentorshipRequestTest – Mentorship Request Lifecycle (23 tests)

Service layer (MentorshipService) + HTTP endpoints; requires MySQL. Ordered tests share state.

ID	Test Method	Description / Input	Expected Result	Type	Status
TC-001	testCreateValidRequest	Student 1 sends request to Mentor 1 (goal 'career', valid message)	HTTP 201; success=true; response has request object with id > 0	Service	**Pass***
TC-002	testPreventDuplicateActiveRequest	Same student sends a second request to same mentor while first is active	409 Conflict; success=false; message contains 'active mentorship request'	Service	**Pass***
TC-003	testPreventSelfRequest	studentId and mentorId are the same user	400; success=false; message contains 'yourself'	Service	**Pass***
TC-004	testPreventRequestToNonAlumni	Request addressed to a user who is a student, not alumni	400; message contains 'alumni mentor'	Service	**Pass***
TC-005	testPreventRequestFromNonStudent	Alumni user submits a request acting as the student	400; message contains 'registered students'	Service	**Pass***
TC-006	testNonExistentStudentRejection	studentId = 9999999	404; message contains 'Student user not found'	Service	**Pass***
TC-007	testNonExistentMentorRejection	mentorId = 9999999	404; message contains 'Mentor user not found'	Service	**Pass***
TC-008	testValidateEmptyMessage	Message is whitespace only (' ')	400; errors object contains 'message'	Validation	**Pass***
TC-009	testValidateShortMessage	Message 'Short' (fewer than 10 characters)	400; errors object contains 'message'	Validation	**Pass***
TC-010	testValidateLongMessage	Message of 1005 characters (limit 1000)	400; errors object contains 'message'	Validation	**Pass***
TC-011	testValidateEmptySessionGoal	sessionGoal is empty string	400; errors object contains 'sessionGoal'	Validation	**Pass***
TC-012	testValidateLongSessionGoal	sessionGoal of 105 characters (limit 100)	400; errors object contains 'sessionGoal'	Validation	**Pass***
TC-013	testGetRequestsByStudent	Fetch requests for Student 1	200; at least 1 request; studentId matches; mentor object present with company 'Google'	Service	**Pass***
TC-014	testGetRequestsByMentor	Fetch requests for Mentor 1	200; at least 1 request; mentorId matches; student object present with department 'Computer Engineering'	Service	**Pass***
TC-015	testMentorAcceptRequest	Mentor 1 sets status ACCEPTED with a response note	200; status = ACCEPTED; mentorResponse saved as entered	Service	**Pass***
TC-016	testUnauthorizedStatusUpdate	A different mentor tries to REJECT the request	403; message contains 'not authorized'	Security	**Pass***
TC-017	testInvalidStatusTransition	Status value 'INVALID_STATUS_STRING'	400 Bad Request	Validation	**Pass***
TC-018	testMentorRejectRequest	Mentor 1 sets status REJECTED with a note	200; status = REJECTED	Service	**Pass***
TC-019	testAllowNewRequestAfterRejection	Student sends a new request after the earlier one was REJECTED	201; success=true (re-request allowed)	Service	**Pass***
TC-020	testMaxMenteeCapacityEnforcement	Mentor with maxMentees=1 accepts Student 1; Student 2 then requests same mentor	Accept returns 200; second request returns 409 with 'maximum mentee capacity'	Service	**Pass***

ID	Test Method	Description / Input	Expected Result	Type	Status
TC-021	testHttpCreateRequestEndpoint	POST /api/requests with student Bearer token and valid JSON	201; success=true; message 'Mentorship request sent successfully!'	API	**Pass***
TC-022	testHttpGetRequestsEndpoint	GET /api/requests?studentId=... with student token	200; success=true; requests array length >= 1	API	**Pass***
TC-023	testHttpPutRequestStatusEndpoint	PUT /api/requests?id=... with mentor token, status ACCEPTED	200; success=true; request status = ACCEPTED	API	**Pass***

3.2 ValidationTest – Input Validation, Registration & Authentication (28 tests)

Pure validator tests plus HTTP, servlet, CGI and TCP socket registration endpoints.

ID	Test Method	Description / Input	Expected Result	Type	Status
TC-024	testValidEmails	student@university.edu, john.doe@company.com, user+tag@domain.co.in	isValidEmail returns true for all	Unit	**Pass***
TC-025	testInvalidEmails	'', 'not-an-email', 'missing@domain', '@nodomain.com'	isValidEmail returns false for all	Unit	**Pass***
TC-026	testValidMobileNumbers	9876543210, +919876543210, 8123456789, 7012345678	isValidMobile returns true for all	Unit	**Pass***
TC-027	testInvalidMobileNumbers	'', '12345', '5123456789' (starts with 5), 'abcdefghij'	isValidMobile returns false for all	Unit	**Pass***
TC-028	testValidPasswords	StrongP@ssw0rd, C0mpl3x#P@ss!, Secure123$Auth	isValidPassword returns true for all	Unit	**Pass***
TC-029	testWeakPasswords	'short', no uppercase, no lowercase, no special char, no number	isValidPassword returns false for all	Unit	**Pass***
TC-030	testValidStudentRegistrationPayload	Complete valid student registration data	ValidationResult.isValid() = true	Unit	**Pass***
TC-031	testStudentRegistrationMissingFields	Empty student payload	Invalid; errors for fullName, email, mobileNumber, password, studentId, department, graduationYear	Unit	**Pass***
TC-032	testStudentRegistrationInvalidFormats	Bad name '1', email, mobile '123', weak password, studentId 'ab', graduation year 1990	Invalid; email message 'Please provide a valid email address.'; errors for password and studentId	Unit	**Pass***
TC-033	testValidAlumniRegistrationPayload	Complete valid alumni data with LinkedIn URL	ValidationResult.isValid() = true	Unit	**Pass***
TC-034	testAlumniRegistrationMissingFields	Alumni payload missing required fields	Invalid; errors for fullName, email, company, designation	Unit	**Pass***
TC-035	testAlumniRegistrationInvalidFields	Graduation year 2045 and invalid LinkedIn profile	Invalid; errors for graduationYear and linkedInProfile	Unit	**Pass***
TC-036	testPostRegisterStudentSuccess	POST /api/auth/register/student with valid data	201; success=true; response contains id	API	**Pass***

ID	Test Method	Description / Input	Expected Result	Type	Status
TC-037	testPostRegisterStudentInvalid	POST student registration with email 'invalid-email'	400; success=false; errors present	API	**Pass***
TC-038	testPostRegisterAlumniSuccess	POST /api/auth/register/alumni with valid data	201; success=true; response contains id	API	**Pass***
TC-039	testPostRegisterAlumniInvalid	POST alumni registration with only a name	400; success=false; errors present	API	**Pass***
TC-040	testUnifiedRegistrationRouting	POST /api/auth/register with role 'student'	201; response role = 'student'	API	**Pass***
TC-041	testServletStyleRegistration	POST /api/servlet/register (JSON, role student)	201; success=true	API	**Pass***
TC-042	testCgiStyleFormRegistration	POST /cgi-bin/register with URL-encoded form body	201; success=true	API	**Pass***
TC-043	testTcpSocketRegistration	Newline-delimited JSON registration sent over TCP socket	Non-null response line; success=true; response has id	Integration	**Pass***
TC-044	testGetRegistrationByEmail	Register alumni, then GET /api/auth/registrations/{email} with token	200; fullName 'Maya Sen', company 'Salesforce'; passwordHash not exposed	API	**Pass***
TC-045	testLoginSuccessAndInvalid	Login with unknown user, then register student and log in correctly	Unknown user: 401, success=false; correct credentials: login succeeds	API	**Pass***
TC-046	testFullNameValidationCases	Valid: 'Aarav Mehta', 'J. Doe', 'Mary-Jane Watson', 'Dr. Bruce Banner'. Invalid: null, '', spaces, 'A', '12345', 'Aarav123', '@Alex!'	Valid names accepted; all invalid names rejected	Unit	**Pass***
TC-047	testPasswordStrengthEvaluator	Evaluate empty, short, and no-uppercase passwords	Empty: score 0 'Very Weak'; short: score 1; no-uppercase: score 2 with correct criterion flags	Unit	**Pass***
TC-048	testHttpPasswordStrengthEndpoint	POST /api/auth/password-strength with 'Sup3r$tr0ngP@ss1'	200; valid=true; level 'Strong'; score 4	API	**Pass***
TC-049	testDuplicateEmailRegistrationFails	Register two students with the same email	First 201; second 409; success=false	API	**Pass***
TC-050	testDuplicateMobileRegistrationFails	Register two students with the same mobile number	First 201; second 409; success=false	API	**Pass***
TC-051	testCheckAvailabilityEndpoint	GET /api/auth/check-availability for a seeded account and for fresh credentials	Seeded: emailAvailable=false, mobileAvailable=false; fresh: both true	API	**Pass***

3.3 SearchAlgorithmsTest – Mentor Search Algorithms (Linear, Quicksort, Binary) (19 tests)

10-mentor in-memory dataset plus HTTP search endpoints on port 5057.

ID	Test Method	Description / Input	Expected Result	Type	Status
TC-052	testManualCompare	Compare ints (1,2), (5,5), (10,2) and strings ('Apple','Google'), ('google','Google'), ('Nvidia','Amazon')	Returns -1, 0, 1 for ints; string compare is case-insensitive and ordered correctly	Unit	**Pass***

ID	Test Method	Description / Input	Expected Result	Type	Status
TC-053	testManualSortStringKey	Manual sort by company, ascending	List sorted; first 'Amazon', last 'Nvidia'	Unit	**Pass***
TC-054	testManualSortNumericalKey	Manual sort by experienceYears, ascending	List sorted; first 2 years, last 7 years	Unit	**Pass***
TC-055	testLinearSearchCompanyMatch	Linear search query 'microsoft'	2 results (Anurag Patil, Vishwesh Bhilare); comparisons = dataset size	Unit	**Pass***
TC-056	testLinearSearchSkillsMatch	Linear search query 'CUDA'	1 result: Aditya Kulkarni	Unit	**Pass***
TC-057	testLinearSearchMultiToken	Linear search query 'Python Cloud'	At least 1 result including Vishwesh Bhilare	Unit	**Pass***
TC-058	testLinearSearchNoMatch	Query 'NonExistentXYZCompany'	0 results; comparisons = dataset size	Unit	**Pass***
TC-059	testLinearSearchEmptyQuery	Empty query string	All records returned; 0 comparisons	Unit	**Pass***
TC-060	testBinarySearchExactMatch	Sort by fullName, binary search 'Rohan Mali'	1 result 'Rohan Mali'; comparisons <= 8	Unit	**Pass***
TC-061	testBinarySearchPrefixMatches	Binary search prefix 'Micro' on company	2 results, both 'Microsoft'	Unit	**Pass***
TC-062	testBinarySearchNumericalGraduationYear	Binary search '2022' on graduationYear	3 results, all graduation year 2022	Unit	**Pass***
TC-063	testBinarySearchNotFound	Binary search 'Tesla' on company	0 results; comparisons <= 5	Unit	**Pass***
TC-064	testBinarySearchEmptyDataset	Binary search on an empty list	0 results; 0 comparisons	Unit	**Pass***
TC-065	testLinearAndBinaryEquivalenceSingleItem	Search 'Google' on company with both algorithms	Same result count, userId and fullName	Unit	**Pass***
TC-066	testLinearAndBinaryEquivalenceMultipleItems	Search 'Microsoft' on company with both algorithms	Both return 2 results with identical ID sets	Unit	**Pass***
TC-067	testHttpSearchLinear	GET /api/mentors/search?query=Microsoft&algorithm=linear	success=true; algorithm 'linear'; metrics totalRecords > 0 and comparisons > 0; at least 2 results	API	**Pass***
TC-068	testHttpSearchBinaryLogarithmic	GET search with algorithm=binary&key=company, query Google	success=true; algorithm 'binary'; searchKey 'company'; at least 1 result	API	**Pass***
TC-069	testHttpSearchFacetFilters	GET search with department=Computer and minExperience=3	Every result department contains 'computer' and experienceYears >= 3	API	**Pass***
TC-070	testHttpGetMentorById	GET /api/mentors, then GET /api/mentors/{id}, then /api/mentors/999999	Valid ID returns matching mentor; unknown ID returns 404	API	**Pass***

3.4 ProfileTest – Profile Management & Avatar (13 tests)

Profile DAO/service tests plus HTTP profile and avatar endpoints on port 5058.

ID	Test Method	Description / Input	Expected Result	Type	Status
TC-071	testGetStudentProfile	Fetch profile of a registered student	Name, role 'student', department, graduationYear 2025 and studentId present	DAO	**Pass***
TC-072	testGetAlumniProfile	Fetch profile of a registered alumnus	Role 'alumni'; company, designation, experienceYears 3, skills as registered	DAO	**Pass***
TC-073	testUpdateStudentProfileSuccess	Update name, mobile, department, graduation year	200; success=true; updated values returned	Service	**Pass***
TC-074	testUpdateAlumniProfileSuccess	Update all alumni fields including LinkedIn URL	200; success=true; all updated fields returned (company Google, maxMentees 5)	Service	**Pass***
TC-075	testMobileNumberCollisionFails	Update student 1 with student 2's mobile number	409; success=false; message contains 'mobile number'	Service	**Pass***
TC-076	testKeepOwnMobileNumberSucceeds	Update profile keeping own existing mobile	200; success=true	Service	**Pass***
TC-077	testValidationFailsOnInvalidName	fullName 'Name123WithNumbers'	400; success=false	Validation	**Pass***
TC-078	testHttpEndpointGetProfile	GET /api/profile?id=... with student token	200; success=true; profile id matches	API	**Pass***
TC-079	testHttpEndpointPutProfile	PUT /api/profile?id=... with alumni token and full payload	200; success=true; company 'Meta'; designation 'Director of Engineering'	API	**Pass***
TC-080	testHttpEndpointNotFound	GET /api/profile?id=9999999 with admin token	404 Not Found	API	**Pass***
TC-081	testAvatarBlobDaoUpdateAndGet	Save and read avatar bytes via DAO (image/jpeg)	Save returns true; bytes and MIME type match; profile avatarUrl = /api/users/{id}/avatar	DAO	**Pass***
TC-082	testHttpAvatarUploadAndStream	POST /api/users/avatar with base64 PNG, then GET /api/users/{id}/avatar	Upload 200; stream 200; content type image/png; bytes identical	API	**Pass***
TC-083	testHttpAvatarDelete	DELETE /api/users/avatar?userId=..., then GET avatar	Delete 200; subsequent GET returns 404	API	**Pass***

3.5 JDBCTest – Database Connectivity & Transactions (4 tests)

Requires a running MySQL instance; test records are cleaned up after each test.

ID	Test Method	Description / Input	Expected Result	Type	Status
TC-084	testDatabaseConnection	Open a connection via DBConnection.getConnection()	Connection non-null, open and valid within 2 seconds	DB	**Pass***
TC-085	testQueryAlumni	Call AlumniDAO.getAllAlumni()	Non-empty list; first record has fullName and company	DB	**Pass***
TC-086	testTransactionalStudentRegistration	Register a student (users + students tables) in one transaction	Positive user ID; existsByEmail and existsByMobileNumber return true	DB	**Pass***
TC-087	testTransactionalAlumniRegistration	Register an alumnus (users + alumni tables) in one transaction	Positive user ID; findById returns company 'JDBC Tech Labs' and designation 'Lead Architect'	DB	**Pass***

3.6 ConflictDetectionTest – Session Schedule Conflict Detection (9 tests)

Pure unit tests of ConflictDetector; no database required.

ID	Test Method	Description / Input	Expected Result	Type	Status
TC-088	testExactMatchConflict	Identical start and end time on the same date (10:00-11:00)	Overlap = true	Unit	**Pass***
TC-089	testPartialOverlapStartsBefore	Candidate 09:30-10:30 vs existing 10:00-11:00	Overlap = true	Unit	**Pass***
TC-090	testPartialOverlapStartsInside	Candidate 10:30-11:30 vs existing 10:00-11:00	Overlap = true	Unit	**Pass***
TC-091	testEnclosingConflict	Candidate 09:00-12:00 encloses existing 10:00-11:00	Overlap = true	Unit	**Pass***
TC-092	testEnclosedInsideConflict	Candidate 10:15-10:45 inside existing 10:00-11:00	Overlap = true	Unit	**Pass***
TC-093	testAdjacentBoundaryBeforeNoConflict	Candidate 09:00-10:00 vs existing 10:00-11:00	Overlap = false (touching boundary is allowed)	Boundary	**Pass***
TC-094	testAdjacentBoundaryAfterNoConflict	Candidate 11:00-12:00 vs existing 10:00-11:00	Overlap = false (touching boundary is allowed)	Boundary	**Pass***
TC-095	testDifferentDateNoConflict	Same time range on two different dates	Overlap = false	Unit	**Pass***
TC-096	testMultiSlotEvaluation	Three existing slots; check 10:00-11:00, 14:30-15:30 and 16:00-15:00 on the first date	First: no conflict; second: conflict with slot ID 2 and message 'Schedule conflict detected'; third: conflict, message contains 'strictly after start time'	Unit	**Pass***

3.7 SessionHistorySortTest – Session History Sorting Algorithms (7 tests)

Manual Bubble, Selection and Insertion sort on four sample sessions; no database required.

ID	Test Method	Description / Input	Expected Result	Type	Status
TC-097	testInsertionSortDurationDesc	Insertion sort by duration, descending	Order 90, 60, 45, 30; comparisons > 0; elapsed nanos > 0	Unit	**Pass***
TC-098	testInsertionSortPriorityDesc	Insertion sort by priority, descending	URGENT, HIGH, MEDIUM, LOW	Unit	**Pass***
TC-099	testSelectionSortSeverityDesc	Selection sort by status severity, descending	FAILED, PAUSED, STARTED, COMPLETED	Unit	**Pass***
TC-100	testBubbleSortChronologicalAsc	Bubble sort by start time, ascending	Session IDs in order 1, 3, 2, 4	Unit	**Pass***
TC-101	testAllAlgorithmsProduceIdenticalOutput	Sort by priority with all three algorithms	Identical ordering at every index	Unit	**Pass***
TC-102	testEdgeCases	Empty list and single-item list	Empty: empty result, 0 comparisons; single: one item (id 10) returned	Boundary	**Pass***
TC-103	testDuplicateKeysHandling	Three sessions with identical duration	All 3 kept; original order 1, 2, 3 preserved (stable)	Boundary	**Pass***

3.8 LoginBoundaryTest – Login Boundary & Security Testing (26 tests)

HTTP tests against POST /api/auth/login covering boundaries, injection and malformed input.

ID	Test Method	Description / Input	Expected Result	Type	Status
TC-104	testValidLoginSuccess	Valid student email and password	200; 'Login successful!'; user role 'student', department 'Computer Engineering'	API	**Pass***
TC-105	testValidAlumniSeedLoginSuccess	Seeded alumni anurag.patil@microsoft.com	200; role 'alumni'; company 'Microsoft'	API	**Pass***
TC-106	testEmptyPayloadBoundary	Empty JSON body {}	400; 'Email and password are required.'	Boundary	**Pass***
TC-107	testEmptyEmailBoundary	Empty email string	400; success=false	Boundary	**Pass***
TC-108	testWhitespaceOnlyEmailBoundary	Email ' '	400; success=false	Boundary	**Pass***
TC-109	testEmptyPasswordBoundary	Empty password	400; success=false	Boundary	**Pass***
TC-110	testMissingEmailField	Payload without email field	400; success=false	Boundary	**Pass***
TC-111	testMissingPasswordField	Payload without password field	400; success=false	Boundary	**Pass***
TC-112	testEmailWithLeadingAndTrailingWhitespace	Valid email padded with spaces	200; email trimmed and login succeeds	Boundary	**Pass***
TC-113	testEmailAllUppercaseBoundary	Valid email in upper case	200; email matching is case-insensitive	Boundary	**Pass***
TC-114	testEmailMixedCaseBoundary	Valid email in mixed case	200; login succeeds	Boundary	**Pass***
TC-115	testPasswordCaseSensitivity	Lower-case variant of correct password	401; 'Invalid email or password.'	Security	**Pass***
TC-116	testPasswordSingleCharacterOff	Correct password with last character changed	401; success=false	Security	**Pass***
TC-117	testSingleCharacterEmailBoundary	Email 'a' (non-existent)	401; success=false	Boundary	**Pass***
TC-118	testMaxValidRfcEmailLengthBoundary	Non-existent email of exactly 254 characters	401; rejected safely, no 500 error	Boundary	**Pass***
TC-119	testOverflowEmailLengthBoundary	Email of 1000+ characters	401 or 400; server does not crash or return 500	Boundary	**Pass***
TC-120	testPasswordUpperLimit128CharsBoundary	Registered user with exact 128-character password	200; success=true	Boundary	**Pass***
TC-121	testExtremeLengthPasswordBoundary	Password of 4000+ characters	401; success=false	Boundary	**Pass***
TC-122	testSqlInjectionTautologyInEmail	Email "' OR '1'='1"	401; authentication not bypassed	Security	**Pass***
TC-123	testSqlInjectionCommentInEmail	Email "admin'--"	401; success=false	Security	**Pass***
TC-124	testSqlInjectionUnionSelectInEmail	UNION SELECT injection string in email	401; PreparedStatement parameterizes safely	Security	**Pass***
TC-125	testSqlInjectionInPassword	Password "' OR '1'='1" with valid email	401; success=false	Security	**Pass***
TC-126	testXssScriptTagInEmail	Email containing a <script> tag	401; success=false	Security	**Pass***
TC-127	testUnicodeAndEmojiBoundary	Email and password containing emoji	401; handled without server failure	Boundary	**Pass***
TC-128	testNullByteBoundary	Email and password containing null bytes	401; success=false	Security	**Pass***
TC-129	testInvalidHttpMethodOnLogin	GET request to /api/auth/login	405 Method Not Allowed	API	**Pass***

3.9 AuthProfileIntegrationTest – Auth & Profile End-to-End Integration (16 tests)

Ordered end-to-end flow: availability check, registration, login, profile, avatar, directory search.

ID	Test Method	Description / Input	Expected Result	Type	Status
TC-130	testStudentStep1_PreRegistrationAvailabilityCheck	GET check-availability for new student email and mobile	200; emailAvailable=true and mobileAvailable=true	Integration	**Pass***
TC-131	testStudentStep2_RegistrationSuccess	Register new student; verify database rows	201; role 'student'; DB row has matching full name and email	Integration	**Pass***
TC-132	testStudentStep3_AvailabilityCheckNowReportsInUse	Repeat availability check after registration	200; emailAvailable=false and mobileAvailable=false	Integration	**Pass***
TC-133	testStudentStep4_LoginWithRegisteredCredentials	Login with the new student's credentials	200; 'Login successful!'; user fields (name, role, department, studentId, graduationYear 2026) match	Integration	**Pass***
TC-134	testStudentStep5_GetProfileMatchesRegistration	GET /api/profile?id=... with student token	200; profile fields match registration data	Integration	**Pass***
TC-135	testStudentStep6_UpdateProfilePersistsChanges	PUT profile with new name and mobile, then re-fetch	200 'Profile updated successfully'; re-fetch shows updated name, mobile, department, year 2027	Integration	**Pass***
TC-136	testStudentStep7_AvatarUploadAndStreaming	Upload base64 avatar, stream it back, log in again	Upload 200 with avatarUrl; GET returns 200 image/png with bytes; login returns avatarUrl	Integration	**Pass***
TC-137	testAlumniStep1_RegistrationSuccess	Register new alumnus with LinkedIn URL	201; role 'alumni'; id returned	Integration	**Pass***
TC-138	testAlumniStep2_LoginSuccess	Login as the new alumnus	200; role 'alumni'; company 'Integration Labs Corp'; designation 'Staff Platform Engineer'	Integration	**Pass***
TC-139	testAlumniStep3_GetProfile	GET alumni profile	200; company, designation, experienceYears 7, maxMentees 4	Integration	**Pass***
TC-140	testAlumniStep4_UpdateProfileAndReflectInDirectorySearch	Update company and designation, then linear search the new company name	Update 200; search finds mentor with company 'Quantum AI Systems', designation 'VP of Engineering', 8 years	Integration	**Pass***
TC-141	testConflict_DuplicateEmailRegistrationFails	Register another student with an existing email	409; errors contains 'email'	Integration	**Pass***
TC-142	testConflict_DuplicateMobileRegistrationFails	Register another student with an existing mobile	409; errors contains 'mobileNumber'	Integration	**Pass***
TC-143	testLoginFailure_WrongPasswordForExistingUser	Existing email with wrong password	401; 'Invalid email or password.'	Integration	**Pass***
TC-144	testProfileUpdateFailure_InvalidMobileLeavesProfileUntouched	PUT profile with mobile '12345'	400; success=false; stored mobile unchanged	Integration	**Pass***
TC-145	testProfileNotFound_NonExistentUserId	GET /api/profile?id=999999999 with admin token	404 Not Found	Integration	**Pass***

Appendix A: Additional Test Cases (TC-146 to TC-169)

These two test classes were in the uploaded files but are not part of the 145 cases above. Delete this appendix if they are out of scope.

A.1 AdminRoleAccessTest – Admin Role Access & RBAC (13 tests)

Not part of the 145-case set; found in the uploaded files. HTTP tests for admin login, role-based access and user management.

ID	Test Method	Description / Input	Expected Result	Type	Status
TC-146	testAdminAuth_SuccessfulLogin	Login as admin@college.edu	200; success=true; token present; role 'admin'	API	**Pass***
TC-147	testAdminAuth_InvalidCredentialsFails	Admin email with wrong password	401; success=false	Security	**Pass***
TC-148	testDuplicateChecking_BatchScan	GET /api/admin/duplicates with admin token	200; response has 'duplicates' and 'totalClusters'	API	**Pass***
TC-149	testDuplicateChecking_ManualSingleCandidateCheck	POST /api/admin/duplicates/check with an existing student email	200; hasDuplicates=true; duplicateCount >= 1; match includes the student	API	**Pass***
TC-150	testRoleRBAC_AdminCanAccessUserDirectory	GET /api/admin/users as admin	200; non-empty users array	Security	**Pass***
TC-151	testRoleRBAC_StudentForbiddenFromAdminUsers	GET /api/admin/users with student token	403 Forbidden	Security	**Pass***
TC-152	testRoleRBAC_AlumniForbiddenFromAdminUsers	GET /api/admin/users with alumni token	403 Forbidden	Security	**Pass***
TC-153	testRoleRBAC_UnauthenticatedForbiddenFromAdminEndpoints	GET /api/admin/users with no token	401 Unauthorized	Security	**Pass***
TC-154	testRoleRBAC_StudentForbiddenFromDuplicatesScan	GET /api/admin/duplicates with student token	403 Forbidden	Security	**Pass***
TC-155	testAdminActions_CreateAndApproveUser	Register a student, then admin approves via POST /api/admin/users/{id}/approve	Approve 200; DB approval_status = 'APPROVED' and is_active = true	Integration	**Pass***
TC-156	testAdminActions_SelfDeleteForbidden	Admin deletes own account	400 Bad Request	Security	**Pass***
TC-157	testAdminActions_StudentForbiddenFromDeletingUser	Student token calls DELETE /api/admin/users/{id}	403 Forbidden	Security	**Pass***
TC-158	testAdminActions_AdminSuccessfullyDeletesUser	Admin deletes a user	200; success=true; user no longer in database	Integration	**Pass***

A.2 SessionFeedbackIntegrationTest – Session History & Feedback (11 tests)

Not part of the 145-case set; found in the uploaded files. Ordered 11-step flow from request to admin feedback review.

ID	Test Method	Description / Input	Expected Result	Type	Status
TC-159	testStep1_CreateMentorshipRequestAndAccept	Student creates request; mentor accepts via PUT	Create 201; GET 200 with request; accept 200	Integration	**Pass***
TC-160	testStep2_StartSessionHistory	POST /api/session-history	201; success=true; status 'STARTED'; session id > 0	Integration	**Pass***
TC-161	testStep3_FeedbackRejectedForIncompleteSession	Submit feedback while session is STARTED	400 Bad Request	Integration	**Pass***
TC-162	testStep4_CompleteSession	PATCH session status to COMPLETED	200	Integration	**Pass***
TC-163	testStep5_FeedbackRatingBoundaryRejection	Feedback with rating 6 and rating 0	Both return 400	Boundary	**Pass***
TC-164	testStep6_FeedbackEmptyCommentsRejection	Feedback with rating 4 and blank comments	400 Bad Request	Validation	**Pass***
TC-165	testStep7_SaveValidFeedbackSuccess	Valid feedback (rating 5, comments) on completed session	201; success=true; DB row matches session, student, mentor, rating, comments	Integration	**Pass***
TC-166	testStep8_DuplicateFeedbackPrevented	Second feedback for the same session	400 Bad Request	Integration	**Pass***
TC-167	testStep9_GetFeedbackBySession	GET /api/feedback/session/{id}	200; feedback fields and student/mentor names match	Integration	**Pass***
TC-168	testStep10_GetMentorFeedbackSummary	GET /api/feedback/mentor/{id}	200; totalReviews 1; averageRating 5.0; distribution 5-star = 1	Integration	**Pass***
TC-169	testStep11_AdminGetAllFeedback	GET /api/feedback with admin token	200; success=true; feedback list has at least 1 entry	Integration	**Pass***

Appendix B: Legend

Type	Meaning
Unit	Tests a single class or method in isolation, with no database or server
Validation	Input validation rules and error messages
Boundary	Edge values: empty, maximum length, adjacency, duplicates
Service / DAO / DB	Business logic, data-access and JDBC tests that need MySQL
API	HTTP, servlet, CGI or socket endpoint tests against the embedded server
Security	Access control, injection and authentication abuse tests
Integration	Multi-step flows spanning several endpoints and the database